On-Device Speech Recognition on iOS: Short Overview

Prepared by BeanBao · Nov 2, 2026

1. Apple's classic API: SFSpeechRecognizer

Apple's Speech framework lets a request be kept on the device. Setting requiresOnDeviceRecognition to true (iOS 13+) stops an SFSpeechRecognitionRequest from sending audio over the network. Apple notes that on-device requests won't be as accurate. The request only honors the setting if the recognizer's supportsOnDeviceRecognition is also true. [5] [6]

Practical pattern (my inference from the docs): check supportsOnDeviceRecognition first, then set requiresOnDeviceRecognition = true. Don't assume the flag alone guarantees on-device processing. Custom language models (WWDC23) are serviced strictly on device, and the app should enforce on-device recognition when it builds the request. [12] [4]

2. Apple's newer API: SpeechAnalyzer / SpeechTranscriber (iOS 26+)

WWDC25 session 277 introduces SpeechAnalyzer as Apple's new speech-to-text API. It is powered by a new Apple-engineered model, and SpeechTranscriber operates on device. DictationTranscriber is the fallback for unsupported languages or devices. Apple's documentation lists SpeechAnalyzer, AssetInventory, SpeechTranscriber and DictationTranscriber as new types. Developers should check SpeechTranscriber.isAvailable or supportedLocales for device support, and manage locale models through AssetInventory. [13] [7] [8] [10]

Architecture: you add module(s) such as a transcriber to a SpeechAnalyzer session, feed it audio, and read results asynchronously. Model retention can be configured to keep resources cached between analyzer instances. [11] [13]

3. Third-party options: WhisperKit and whisper.cpp

WhisperKit (Argmax, open-source Swift package, now in the argmax-oss-swift repository) runs Whisper-style models on device and automatically downloads a recommended model for the device. Its authors' workshop paper reports 0.46 s latency and 2.2% WER. These are vendor-reported numbers, not an independent evaluation. [14] [1]

whisper.cpp is a C/C++ port of OpenAI's Whisper with Apple Silicon optimizations (Metal, Core ML), and it ships iOS example apps. Its README claims a Core ML/Neural Engine encoder speed-up of more than 3x over CPU-only. That is the project's own claim, and you must generate the Core ML model yourself. [16] [15]

4. Caveats and uncertainty (secondary sources)

Apple's pages I retrieved don't list which languages work on-device or when models download. Community reports (unverified) say offline dictation can fail, for example with error 1101, unless the

keyboard and dictation language are installed. One blog says a language model arrives only after the system fetches it, so "supported" doesn't guarantee "installed". A 2022 Medium post found far fewer on-device languages than server mode, and it may be outdated. [9] [3] [18]

Accuracy and speed comparisons between SpeechAnalyzer and Whisper come from third parties. One hands-on test reported a roughly 2.2x speed advantage over Whisper Large V3 Turbo. Argmax is a competing vendor and publishes its own table. Treat all of these as anecdotal, because Apple has published no accuracy figures I could find. A blog claim that SpeechTranscriber doesn't accept contextual strings is also unconfirmed. [17] [19] [2]

Sources

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